Check for updates

Received: 27 February 2025 | Revised: 25 July 2025 | Accepted: 8 August 2025

DOI: 10.1111/medu.70030

RESEARCH

Breaking the silence: Revealing drivers and barriers to medical students' speaking up in medical error

Jen-Chieh Wu[1,2] | Hung-Chen Chen[3] | Enoch Yi-No Kang[4,5,6] | Hung-Wei Tsai[1,2] | Yi-Chun Chen[7] | Hao-Yu Chen[3] | Hui-Wen Chen[8] | S. Barry Issenberg[9]

[1]Department of Education and Humanities in Medicine, School of Medicine, College of Medicine, Taipei Medical University, Taipei, Taiwan

[2]Department of Emergency Medicine, Taipei Medical University Hospital, Taipei, Taiwan

[3]Center for Education in Medical Simulation, Taipei Medical University, Taipei, Taiwan

[4]Institute of Health Policy & Management, College of Public Health, National Taiwan University, Taipei, Taiwan

[5]Department of Health Care Management, College of Health Technology, National Taipei University of Nursing and Health Sciences, Taipei, Taiwan

[6]Center of Medical Education Research, Wan Fang Hospital, Taipei Medical University, Taipei, Taiwan

[7]Department of Emergency Medicine, National Taiwan University Hospital, Yunlin, Taiwan

[8]School of Nursing, College of Nursing, Taipei Medical University, Taipei, Taiwan

[9]University of Miami Gordon Center for Simulation and Innovation in Medical Education, Miami, Florida, USA

Correspondence

Hao-Yu Chen, Center for Education in Medical Simulation, Taipei Medical University, 250 Wuxing Street, Taipei 110301, Taiwan.
Email: haoyu68@tmu.edu.tw

Hui-Wen Chen, School of Nursing, College of Nursing, Taipei Medical University, 250 Wuxing Street, Taipei 110301, Taiwan.
Email: hwchen01@tmu.edu.tw

Funding information

National Science and Technology Council, Grant/Award Number: 113-2410-H-038-016-MY2

Abstract

Introduction: Speaking up about medical errors is a critical behaviour for medical students, as it plays a vital role in enhancing patient safety. Few studies have explored the drivers and barriers affecting their willingness to speak up in clinical training, particularly within hierarchical Asian cultures. The purpose of this study was to explore drivers and barriers shaping medical students' speaking up behaviours about medical errors to inform education, mentorship and patient safety practices.

Methods: This qualitative study using a descriptive phenomenological approach was conducted at a teaching hospital in Northern Taiwan between October 2023 and April 2024. Sixth-year medical students, with 20 months of clinical rotations and simulated speaking-up training, were invited to participate in semi-structured interviews. The interview process was concluded after the 10th participant, as data saturation had been achieved. Data were analysed using thematic analysis to explore factors influencing students' willingness and behaviour to speak up in clinical settings.

Results: Three main themes were observed: (1) *predisposing features*, such as personal traits and prior experiences, which may influence students' speaking up behaviour; (2) *psychological processes*, highlighting the role of professional knowledge, attitudes towards clinical work and ethical and legal considerations; and (3) *contextual interaction*, emphasising the influence of patient urgency, supervisor characteristics and team atmosphere. The research team also found that the themes are interconnected and collectively influence novices' speaking-up behaviours when faced with a medical error.

Conclusion: Confidence and responsibility promote speaking up, but barriers such as hierarchy and fear of criticism hinder it. Although ethical and legal training is designed to promote patient advocacy, it may unintentionally prioritise self-preservation, thereby discouraging speaking-up behaviour in medical error events. These findings may prompt medical educators to re-evaluate the hidden curriculum within ethical and legal training as well as clinical rotations.

Hui-Wen Chen and Hao-Yu Chen contributed to the study equally.

© 2025 Association for the Study of Medical Education and John Wiley & Sons Ltd.

1 | INTRODUCTION

As future members of the health care team, medical students play a crucial role in promoting patient safety by speaking up when they observe medical errors [1, 2]; thus, gaining a deeper understanding of the drivers and barriers that influence their willingness to speak up for medical error is essential. Medical errors, including wrong-site surgeries, diagnostic errors and medication-related incidents [3], affect roughly one in 10 health care encounters and cause over three million deaths annually [4], with more than half of these incidents being preventable. These include medication-related incidents [5]. Accordingly, recognising and addressing medical error is both a critical clinical skill and a core aspect of medical professionalism [6]. However, addressing lapses in professionalism observed in others often poses a significant emotional and hierarchical challenge for medical students or trainees [7]. Therefore, exploring and understanding medical students' perspectives on speaking up about medical errors is essential for preparing them with assertive communication skills through well-designed curricula, targeted behavioural development, mentorship and the cultivation of a supportive clinical culture and policy framework [2, 8–10].

Effectively expressing concerns within a hierarchical structure requires formal training that develops students' emotional intelligence and communication skills [11]. Speaking up is a complex form of communication shaped by various factors [12], including personal and interpersonal conditions [13]. Studies indicate that medical students often display an unwholesome level of submissiveness towards senior doctors [14]. The perceived barriers for medical students to speak up consistently vary depending on the specific characteristics of the medical error situation [15], and include hierarchies and power differentials, seniority and unit tenure, fear of negative feedback or being wrong and concern about damaging relationships [16, 17]. Medical students are more likely to speak up about medical errors when they receive encouragement [18], when they feel confident in their knowledge and communication skills [19], or when the patient is in a life-threatening condition [20]. Thus, formal training, particularly personalised instruction in assertive communication skills, is vital for medical students [13].

In addition to the formal curriculum, clinical events in a hidden curriculum profoundly shape medical students' internalisation of professional norms, values and expectations [21–23]. Embedded in daily interactions and institutional culture, it conveys implicit messages about power, hierarchy and appropriate behaviour in clinical practice [24–26]. While formal education emphasises patient advocacy and encourages speaking up, the hidden curriculum may contribute to silence, deference to authority and conformity [27]. These unspoken norms can lead to hesitation, particularly in ethically complex or legally sensitive situations such as medical errors, where hierarchical dynamics suppress open communication [28]. Recognising hidden curriculum influences, such as structure and culture [23, 26, 29, 30], is essential for interpreting student behaviour and designing training that fosters ethical responsibility and professional courage.

An in-depth understanding of the intricate relationships among medical students' personal attributes and environmental influences on their willingness and behaviour to speak up is essential for developing effective strategies and interventions to enhance safety voice practices [31]. Therefore, the purpose of this study was to explore medical students' perspectives on medical errors, their willingness to speak up, and their speaking-up behaviours, using a phenomenological approach, in order to identify the potential influences that could shape these behaviours and inform the development of educational interventions, mentorship strategies and clinical practices that promote patient safety and professional growth.

2 | METHODS

2.1 | Study designs

This study was conducted at a teaching hospital affiliated with a medical university in northern Taiwan between October 2023 and April 2024. This study used a descriptive phenomenological approach to explore medical students' views on medical errors, willingness to speak up, decision-making processes and related behaviours by examining their conscious perceptions and interpretations of their lived experiences through in-depth semi-structured interviews [32]. Descriptive phenomenology was selected because it allowed the research team to explore the complex influences on speaking-up behaviour by bracketing preconceptions and carefully analysing medical students' lived experiences and subjective perspectives to uncover essential meanings and achieve a deeper understanding [32, 33].

2.2 | Settings and interviewees

This study used purposive sampling [34] to interview sixth-year medical students who had completed 20 months of clinical rotations in key departments, internal medicine, surgery, obstetrics and gynaecology, paediatrics and emergency medicine, as well as coursework in patient safety and medical humanities. At the time of the interviews, they were in the final 3 to 4 months of their 6-year medical programme. By this stage, they were expected to have acquired core clinical competencies, including independently conducting patient interviews and physical examinations, documenting medical records, managing basic medical orders, performing nasogastric and urinary catheterisations and participating in clinical resuscitation efforts, among other essential clinical skills. This population, which has not yet formed strong attitudes in the current clinical context, represents a prime target for assertive communication training.

To ensure participants shared a relevant experiential basis, the research team employed purposive sampling, selecting only medical students who had completed a structured simulation course on speaking up in medical error scenarios [20]. Two weeks prior to the course, students received preparatory materials, including the American Heart Association and Paroxysmal Supraventricular Tachycardia (PSVT) guidelines [35]. The 150-min course began with a conceptual review and quiz, followed by two high-fidelity simulation scenarios—one

13652923, 0, Downloaded from https://asmepublications.onlinelibrary.wiley.com/doi/10.1111/medu.70030 by Lendis Wu - Taipei Medical University(Tmu) , Wiley Online Library on [10/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

involving stable PSVT and the other unstable—incorporating the use of standardised nurses and mannequins. A physician intentionally administered incorrect medication to elicit student responses. Faculty conducted debriefing sessions between simulations to promote reflection on clinical decisions, emotional responses and communication strategies. The application of purposive sampling to the medical students who completed the course guaranteed that all interviewees, including those without direct clinical exposure to medical errors, were able to engage and reflect on the questions in the interview.

A course administrator assisted the research team by announcing the study and inviting medical students to voluntarily participate as interviewees. Interviewees were informed about the study's purpose, and the informed consent process emphasised that their data would remain anonymous, their identities would not be disclosed to the teaching team and their responses would not affect their grades. Additionally, none of their supervisors who had a role in grading them in other clinical settings were involved in the study.

2.3 | Data collection

Data collection was conducted by a research team member with expertise in qualitative research. The interviewer, who had a decade of administrative experience supporting medical student clinical rotation programmes and 7 years of involvement in speaking up training courses, had a comprehensive understanding of speaking up in the context of medical errors and the unique dynamics of medical students and their clinical environments. The interviewer's independence from the speaking-up course's design, delivery and evaluation, coupled with the absence of supervisory or teaching roles over participants, minimised power differentials. This independence reduced bias that may stem from hierarchical pressure, thereby fostering more open disclosure and enhancing data credibility.

Based on literature and observations [36], a flexible interview guide was developed to allow adaptive follow-up questions and encourage deeper exploration of participants' experiences. Interviews began by inviting interviewees to reflect on their course experiences and then explored their decision-making and reasoning in simulated and clinical scenarios. Questions focused on decision-making processes, perceptions and emotional responses to medical errors. No evaluative feedback was provided, reducing social desirability bias and promoting authentic, experience-based reflections and insights. The interview questions were included in Table 1. To ensure consistency in data collection, the interviewer conducted one-on-one interviews within 1 week after the students completed the speaking-up simulation course.

Interviews were audio-recorded in Mandarin, transcribed verbatim using Microsoft Word and Excel, and validated by researchers through repeated listening to ensure accuracy and reliability. The interviewer conducted a preliminary analysis after each interview with another research team member. As three core themes gradually emerged, no new insights appeared by the eighth interview, and the final two interviews confirmed consistency, indicating theoretical saturation [37]. This was supported by the concept of information power, which considers study aim, participant relevance and data quality [38, 39]. As a result, data collection concluded after the 10th participant. Finally, selected quotations were translated into English by one team member, with a subsequent back-translation performed by another. AI-assisted tools (ChatGPT-4o, OpenAI) were utilised to ensure lexical consistency, and a native English speaker reviewed the final English text. All discrepancies were resolved in regular team meetings.

TABLE 1 The interview questions.

1.	Can you recall the content of the teaching plan? What actions did you take, and why did you choose to act that way?
2.	How do you feel about your performance at that time? (Was it calm or emotional?)
3.	What factors do you think influence your decision to speak up or remain silent?
4.	Have you encountered similar experiences during your clinical rotations?
5.	Do you think hierarchy or specialty choice has an impact?
6.	Thank you for sharing. Do you have any additional thoughts or opinions you'd like to share with me?

Note: The interview guide served as a flexible framework rather than a fixed script. Interviews began with open-ended questions, followed by responsive probing based on interviewees' answers.

2.4 | Data analysis

This study employed thematic analysis as its research method, which focuses on identifying key themes from textual data to gain a deeper understanding of interviewees' experiences, perspectives and emotions [36]. The chosen method is well suited for its capacity to systematically uncover and analyse recurring patterns embedded within the intricate and subjective accounts of medical error reporting provided by the interviewees. The analysis followed the six main steps of thematic analysis: (1) familiarisation with the data, (2) generating initial codes, (3) searching for themes, (4) reviewing themes, (5) defining and naming themes and (6) writing the report.

To minimise potential bias and enhance the clinical and educational relevance of the findings, the multidisciplinary research team, comprising a clinical physician, a nursing researcher, a medical education expert and a qualitative methodologist, held biweekly meetings from July to December 2024. During these meetings, the team reviewed interview transcripts, engaged in open reflection on the development and refinement of themes, and ensured transparency and traceability throughout the analytical process. Discussions and meeting notes served as documentation to support interpretive neutrality and thematic coherence. The trustworthiness of this study was evaluated and established using criteria such as transparency and clarity, reflexivity, data saturation and result verification [36].

13652923, 0, Downloaded from https://asmepublications.onlinelibrary.wiley.com/doi/10.1111/medu.70030 by Lendis Wu - Taipei Medical University(Tmu) , Wiley Online Library on [10/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

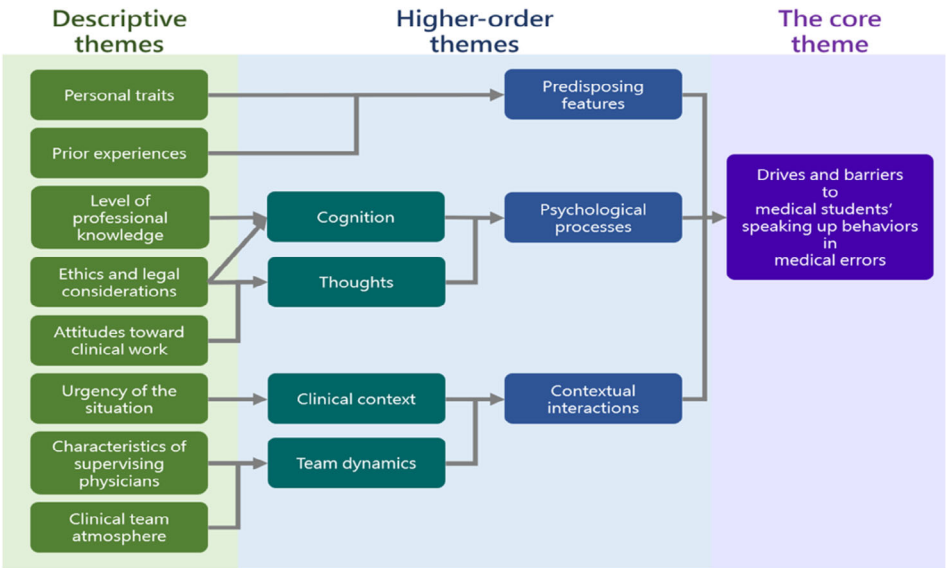

FIGURE 1 Theme maps showing the hierarchy of thematic analysis, including descriptive themes, higher order themes and the core theme.

2.5 | Ethical considerations

This study was approved by the institutional review board of a medical university. The entire research process adhered to ethical standards. Interviewees' personal information was kept private, and they could withdraw anytime. Data were securely stored to protect privacy, ensuring interviewees' rights and well-being are maintained. No adverse reactions were observed among the interviewees involved in the study.

3 | RESULTS

Ten medical students participated in the interviews, consisting of five females and five males, with an average age of 24.4 years (range: 23–29 years). Among them, six students (60%) held roles as class or group leaders. The average interview duration was 51.3 min. Interviews with females lasted about 65 min on average (median 54 min, with the first quartile at 50 min and the third quartile at 84 min), while interviews with males averaged 41.6 min (median 38 min, with the first quartile at 36 min and the third quartile at 46 min). Seven students reported that they experienced medical error events, of whom two personally spoke up, while three observed others speaking up during the incidents.

Through the thematic analysis, eight descriptive themes were identified: (a) 'Personal traits', (b) 'Prior experiences', (c) 'Level of professional knowledge', (d) 'Ethics and legal considerations', (e) 'Attitudes toward clinical work', (f) 'Urgency of the situation', (g) 'Characteristics of supervising physicians' and (h) 'Clinical team atmosphere'. These themes can be categorised into higher order themes such as 'Cognition', 'Thoughts', 'Clinical context', 'Team dynamics', 'Predisposing features', 'Psychological processes' and 'Contextual interactions'. These themes reflect the contextual framework and the manifestation of the phenomenon (Figure 1).

The drivers and barriers influencing students' willingness to speak up are associated with their personal traits, life experiences during upbringing and the development of their thought processes over time. When faced with medical errors, their decisions to speak up are ultimately shaped by the interaction between the surrounding context and their immediate thoughts.

3.1 | Theme 1: Predisposing features

Most of the interviewed medical students identified strengths and weaknesses rooted in their personal traits and prior clinical experiences. These factors shaped their varying perspectives on medical errors, influencing their intuitive or reflexive decisions about whether to speak up when faced with erroneous medical instructions from authoritative physicians in clinical settings. This theme was further divided into two descriptive themes: 'Personal traits' and 'Influence of *prior* experiences', detailed below.

3.1.1 | Descriptive Theme 1.1: Personal traits

During the interviews, medical students noted that their willingness to speak up was influenced by personal traits developed over time. Some described themselves as naturally inquisitive, having been encouraged to ask questions from a young age, while others were taught to prioritise obedience, believing it to be a safer and more harmonious approach. For instance, the following statements were shared by Interviewees I and H. A student who tends to speak up shared:

13652923, 0, Downloaded from https://asmepublications.onlinelibrary.wiley.com/doi/10.1111/medu.70030 by Lendis Wu - Taipei Medical University(Tmu) , Wiley Online Library on [10/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

> Since I'm the kind of person who tends to ask questions, I would try to ask the attending physician about the purpose behind their action, what their thinking process was, and why they were thinking of doing it that way.
>
> (Interviewee I)

Another student who tends not to speak up explained:

> Maybe it's because my mom always told me to be good and obedient when I was little. I think that path—being a good kid—worked out well for me. Everything went smoothly, and I never got into trouble, so I tend to trust the attending physician's decisions first.
>
> (Interviewee H)

3.1.2 | Descriptive Theme 1.2: Influence of prior experiences

Many of the interviewed medical students reported encountering situations during their clinical rotations where they had to decide whether to speak up. The outcomes of these past experiences indirectly influenced their willingness to speak up in future scenarios. For instance, the following statements were shared by Interviewees B and J. A student who had a negative experience after speaking up chose not to speak up again:

> There was a time when a classmate suggested a different idea, and the attending physician questioned where they got that information. Then, you had to provide evidence, or the attending physician might say, 'Why don't you take the lead instead?'
>
> (Interviewee B)

Another student who had a positive experience with their suggestion being accepted chose to speak up:

> For example, with diabetes medications, there are various mechanisms of action, and one of them seems to have a certain side effect. I remembered another drug that might be better. So, I asked the attending physician if that drug didn't have those side effects and if it might be a better choice. The attending physician said it sounded like a good idea and accepted my suggestion.
>
> (Interviewee J)

3.2 | Theme 2: Psychological processes

Clinical work is inherently team-based, with most roles in clinical settings involving collaboration among multiple individuals. Students' willingness to speak up is often indirectly influenced by their perceptions of the broader environment, including the people and situations around them. This aspect can be categorised into three descriptive themes: 'Level of professional knowledge', 'ethics and legal considerations' and 'Attitudes toward clinical work', as described below.

3.2.1 | Descriptive Theme 2.1: Level of professional knowledge

Medical students noted that confidence in their professional knowledge affects their decision to speak up. For instance, the following statements were shared by Interviewees C and F:

> If I don't feel very confident about clinical medicine, I start thinking maybe I haven't studied enough, and then I get really worried that what I say might be the wrong answer.
>
> (Interviewee C)

> You need to have your own knowledge combined with some clinical experience, and only then can you confidently bring up the issue and argue your case.
>
> (Interviewee F)

3.2.2 | Descriptive Theme 2.2: Ethics and legal considerations

Some students highlighted that, even when dealing with an authoritative physician, there are effective ways to communicate. They noted that adopting a friendly attitude and choosing the right timing can lead to better outcomes when speaking up. This consideration of interaction aligns with ethical standards regarding the teacher–student relationship.

> I don't think anyone is so unreasonable that they can't be communicated with, even if they are very authoritative or strict. It's important to consider that the patient might be listening, so how you say things matters even more. You should never contradict the attending physician's decision in front of the patient. Instead, wait until you're at the nurse's station and then talk to the attending physician in a very tactful way, keeping their feelings in mind.
>
> (Interviewee A)

Alternatively, drawing on their training in medical law and regulations, some students indicated that because clinical responsibilities do not directly fall on them, they tend to hold back from speaking up and leave the attending physician accountable for any medical errors. For instance, the following statements were shared by Interviewees B and G:

13652923, 0, Downloaded from https://asmepublications.onlinelibrary.wiley.com/doi/10.1111/medu.70030 by Lendis Wu - Taipei Medical University(Tmu) , Wiley Online Library on [10/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

> I thought it might be better not to stand out. After all, if something goes wrong, it's their (the attending physician's) problem to deal with.
>
> (Interviewee B)

> We are rarely mentioned in the nursing records, so the responsibility is relatively less on us. It makes us feel like it's fine to just let them make the decisions.
>
> (Interviewee G)

3.2.3 | Descriptive Theme 2.3: Attitudes towards clinical work

Although a solid understanding of professional knowledge can increase students' confidence in speaking up about medical errors—since they have a clear sense of what is right in a given clinical situation—their attitudes towards clinical work play a crucial role in influencing this behaviour. Some students with a proactive attitude towards clinical work believe they are doing the right thing and feel that their opinions can positively influence their supervisors while also benefiting the patients.

> I think as long as we keep the patient in mind and use an inquiring tone, the attending physicians are usually receptive. They may even feel that having more people on the team to double-check things is beneficial.
>
> (Interviewee A)

Conversely, some students gradually realise during clinical rotations that their ability to influence the environment is limited. They feel that senior personnel are less likely to change and, as a result, become reluctant to challenge the system.

> You start to realize that health care rarely aligns with the ideal situation you had in mind, and the senior staff are definitely more deeply involved. From our perspective, we only see small ripples. It feels like the more senior they are, the more indifferent they seem.
>
> (Interviewee G)

3.3 | Theme 3: Contextual interactions

The dynamic nature of clinical work and the urgency of events in clinical situations significantly affect students' willingness to speak up. Students highlighted that various roles and interactions within clinical settings also influence their decisions to voice concerns at any given moment. This theme can be further divided into three descriptive themes: 'Urgency of the situation', 'Characteristics of supervising physicians' and 'Clinical team atmosphere', described below.

3.3.1 | Descriptive Theme 3.1: Urgency of the situation

The urgency of a patient's condition plays a crucial role in determining both clinical team communication in general, as well as in shaping students' decision to speak up. When the situation is stable, students feel they have more time to evaluate the appropriate course of action and discuss it with their supervisors.

> I think the most important factor for me is whether the patient is unstable. If the patient is stable, I'll have time to discuss with the attending physician how to proceed. But if the patient is unstable, I'll let the attending physician handle it first.
>
> (Interviewee D)

However, some students noted that if the patient's condition is stable and an incorrect decision would not impact vital signs, they might choose not to speak up. For instance, interviewee A shared that:

> It depends on the patient's condition. If it's not life-threatening, I might not intervene. For minor errors, like prescribing harmless medication, it's not a big deal.
>
> (Interviewee A)

3.3.2 | Descriptive Theme 3.2: Characteristics of supervising physicians

The perceived personality of supervising physicians also influences students' willingness to speak up. Students are less likely to speak up if the supervisor appears stern or unapproachable.

> The attending physician's attitude makes a difference. In the first scenario, I felt like the attending physician seemed very cold and unapproachable, while in the second scenario, the attending physician was more open to discussion. Since the patient in the second scenario was also stable, I felt more comfortable expressing my opinions.
>
> (Interviewee D)

Additionally, if the supervisor is a renowned physician or an experienced senior, students often trust their decisions without question.

> I mean, if they're one of those renowned, authoritative doctors, you're even more likely to believe that they must be right.
>
> (Interviewee H)

13652923, 0, Downloaded from https://asmepublications.onlinelibrary.wiley.com/doi/10.1111/medu.70030 by Lendis Wu - Taipei Medical University(Tmu) , Wiley Online Library on [10/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

3.3.3 | Descriptive Theme 3.3: Clinical team atmosphere

The atmosphere within a clinical department significantly impacts students' sense of burden and pressure when deciding whether to speak up. A positive and supportive environment fosters students' willingness to voice their concerns.

> I think the atmosphere of a department is really important. As someone with no experience, I constantly wonder if asking questions will get me scolded or make me feel uncomfortable. That kind of stress can be a significant burden.
>
> (Interviewee E)

> If the work environment is friendly and you notice that offering your opinion doesn't lead to criticism, disdain, or being ignored, you'll feel increasingly grateful and encouraged to speak up.
>
> (Interviewee G)

3.4 | Relationship among the themes

Based on the descriptions in the main themes, Figure 2 illustrates their conceptual relationships, highlighting how the themes interact with one another. We consistently found that interviewees' personal traits and prior experiences shaped their intention to speak up, particularly their attitudes towards clinical work. Additionally, their knowledge base influences their confidence in performing tasks and interacting with other health care professionals. However, the decision to speak up is further complicated by other factors such as the patient's situation, the characteristics of supervising physicians, and the team atmosphere. Our interviews did not find that predisposing features directly worked on contextual interactions or speaking up behaviours in medical errors. Instead, the responses revealed no clear evidence that these features functioned in either theme of contextual interaction or speaking up behaviour, suggesting they might not play a dramatic role.

4 | DISCUSSION

The findings reveal that medical students' speaking-up behaviour is influenced by personal traits, psychological processes, complex interactions in clinical context, and their sense of responsibility. Proactive attitudes, confidence in their knowledge, positive experiences and a strong sense of responsibility for patient safety encouraged speaking up, particularly in stable scenarios or within supportive teams. Conversely, hierarchical barriers, unapproachable supervisors, fear of criticism and negative past experiences, such as being dismissed, discouraged speaking up. Additionally, students who felt less accountable for clinical outcomes were also less likely to speak up, often deferring responsibility to senior physicians. These findings highlight the need to build students' confidence in knowledge and communication skills, address hierarchical challenges, foster supportive environments and cultivate a sense of shared responsibility to promote speaking-up behaviours.

Our findings corroborate prior research and address existing knowledge gaps. Cognition's role as a key driver of speaking-up behaviour is expected, given that medical students develop professional knowledge, communication skills and clinical experience through various training programmes [40, 41]. This foundational knowledge, in turn, fosters confidence, which is strongly linked to their propensity to speak up about medical errors [12, 13]. However, students with higher academic levels often hesitate to speak up and withhold their voice despite their ability to identify patient safety concerns or rule violations [42]. This indicates the need for further investigation into how clinical experience influences speaking-up behaviours and the role of supportive environments in fostering such

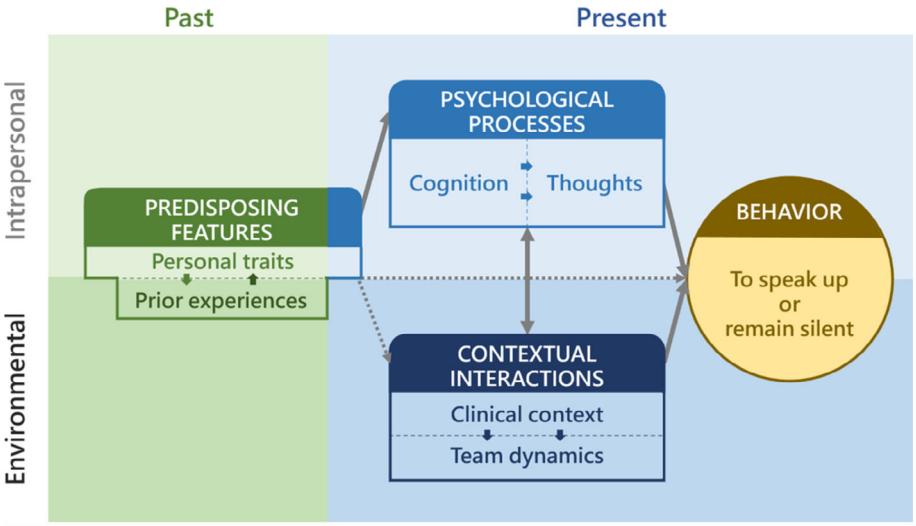

FIGURE 2 Conceptual relationships of drives and barriers to medical students' speaking up behaviours in medical errors.

13652923, 0, Downloaded from https://asmepublications.onlinelibrary.wiley.com/doi/10.1111/medu.70030 by Lendis Wu - Taipei Medical University(Tmu) , Wiley Online Library on [10/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

actions. The environments, as revealed by our findings, appear to integrate both the implicit learning of the hidden curriculum in clinical practice and the explicit instruction of formal training delivered in school and hospital contexts.

4.1 | Psychological safety culture: Invisible but crucial

Our findings emphasise the importance of creating a safe and encouraging atmosphere to break negative behavioural patterns and reinforce positive ones. Medical students who witness errors but feel powerless to intervene may experience psychological distress and moral conflict, becoming 'second victims' in a broader sense due to the lasting emotional and professional consequences [43], even though the term traditionally refers to those directly involved in causing patient harm [44]. From this perspective, speaking up serves not only to safeguard patient safety and support senior colleagues who may have made mistakes but also to help students maintain their own ethical integrity and psychological well-being. This highlights the urgency of implementing training programmes that enhance patient safety and provider resilience [44].

Students who previously withheld their voice often maintained this behaviour, while those with positive prior experiences or confidence gained from earlier attempts were more likely to speak up again [12, 13, 42]. Additionally, while medical students recognise the importance of knowledge in speaking up, they often overlook the critical role of communication skills. Effectively addressing medical errors in hierarchical clinical environments requires not only confidence in knowledge but also strong communication skills to navigate authority gradients and facilitate constructive dialogue [11]. This highlights the need to raise awareness of assertive communication and prioritise its integration into medical curricula, and such programmes must be tailored to meet the needs of both junior and senior health care providers, equipping them to effectively manage and respond to medical errors [11]. However, many current interventions fail to incorporate the latest evidence, limiting their impact [45]. By developing evidence-based training initiatives that emphasise communication skills, educators and researchers can foster a culture of safety and resilience, ultimately benefiting both patients and health care providers. To support such positive behaviours and relevant training environments, future interventions may also involve faculty development initiatives that train supervising physicians to foster psychologically safe environments and actively encourage learner voice in clinical settings [25].

4.2 | Ethics and law in medicine training: A double-edged sword

Medical ethics play a vital role in fostering effective health care teams by promoting respect, collaboration and shared responsibility [46]. However, these principles can inadvertently hinder speaking-up behaviour in the face of medical errors. Ethical obligations to maintain professional relationships and defer to authority can create tension, particularly in hierarchical clinical environments. Medical students may prioritise team harmony over raising concerns about potential errors, fearing that doing so might disrupt relationships or challenge senior physicians [12, 13]. Similarly, while education on medical legal issues enhances students' awareness of responsibilities and the legal implications of errors [47], it may unintentionally discourage speaking up. Students may become overly cautious, focusing on avoiding liability or legal entanglements instead of actively intervening to protect patients [48]. This shift in focus risks compromising their ability to act decisively, prioritising self-preservation over patient advocacy. These behavioural patterns reflect the latent influence of the hidden curriculum, which conveys implicit norms and expectations through everyday clinical routines and hierarchical relationships [24]. Several medical students in our study also mentioned that their limited clinical team status, lacking documentation or decision-making roles, diminished their legitimacy and willingness to speak up.

To address these challenges, educational programmes must reform medical training to include a comprehensive, theory-based and longitudinal approach that emphasises teamwork, communication, evidence-based practice and strategies for managing uncertainty [9]. To support such reform, models such as Longitudinal Integrated Clerkships (LICs) provide a promising framework. By promoting sustained engagement with patients, supervisors and teams, LICs enable students to assume authentic clinical roles and progressively strengthen their professional identity and confidence [49, 50]. Evidence also suggests that LICs may reduce ethical erosion by fostering continuity and reflective learning [51]. Complementary models such as communities of practice [52] and value-added roles [53] further support students' transition from peripheral observers to active contributors. Together, these approaches provide the structural and experiential foundations to embed speaking up as a normalised aspect of professional behaviour, rather than an exceptional or risky act. Ethical and legal training should stress speaking up as a fundamental responsibility, balancing personal risk with the commitment to ensuring patient safety and equipping future doctors to prevent, manage and reduce avoidable errors.

5 | LIMITATIONS

Despite reaching data saturation, the study's transferability is limited by its single-site design at a teaching hospital in Northern Taiwan, where many interviewees held leadership roles, potentially introducing self-selection bias and limiting the diversity of perspectives applicable to broader cultural and educational contexts in other regions. Efforts were made to enhance reliability by having a research team member without a teacher-student relationship conduct the interviews. Our aim was to elicit rich and experience-based insights rather than statistical generalisability. The reliability of themes related to personal traits and prior experiences may still be affected by recall bias. Although all interviewees underwent the same simulation, real clinical

13652923, 0, Downloaded from https://asmepublications.onlinelibrary.wiley.com/doi/10.1111/medu.70030 by Lendis Wu - Taipei Medical University(Tmu) , Wiley Online Library on [10/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

settings differ in culture, supervision and patient acuity, influencing speaking-up behaviour and safety perceptions. Interviewees' reflections often merged simulated and real experiences, enriching authenticity but limiting clarity about context. This is an important consideration when interpreting these findings beyond the study's scope. This issue arises from the absence of relevant documents to verify the information and the lack of long-term follow-up. Therefore, further research is needed to systematically gather relevant documents; conducting long-term follow-up with certain cases would be beneficial to address this issue.

6 | CONCLUSION

Medical students' willingness to speak up about medical errors is shaped by personal traits, professional knowledge, clinical context and personal cognition and thoughts. As medical educators, current observations guide us to rethink the hidden curriculum in both ethical and legal training and clinical rotation. Ethical and legal training is commonly intended to foster patient advocacy, but it may unintentionally shift medical students' focus on self-preservation, potentially hindering open communication in clinical settings. This presents an important opportunity to critically reflect not only on organisational and team cultures but also on the design, content and impact of ethical and legal training programs. By reassessing these elements, we can better align training objectives with the ultimate goal of enhancing patient care and promoting a supportive, transparent work environment.

AUTHOR CONTRIBUTIONS

Jen-Chieh Wu: Project administration; conceptualization; methodology; writing—original draft; writing—review and editing; funding acquisition; resources; validation. **Hung-Chen Chen**: Conceptualization; methodology; writing—original draft; formal analysis; writing—review and editing. **Enoch Yi-No Kang**: Conceptualization; methodology; visualization; writing—review and editing; validation. **Hung-Wei Tsai**: Data curation; methodology; investigation; writing—review and editing. **Yi-Chun Chen**: Data curation; methodology; investigation; writing—review and editing. **Hao-Yu Chen**: Conceptualization; methodology; formal analysis; writing—review and editing. **Hui-Wen Chen**: Conceptualization; methodology; writing—original draft; writing—review and editing; validation. **S. Barry Issenberg**: Conceptualization; supervision; writing—review and editing; validation.

ACKNOWLEDGEMENTS

The authors have nothing to report.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ETHICS STATEMENT

This study was approved by the institutional review board of a medical university.

ORCID

Jen-Chieh Wu https://orcid.org/0000-0003-3606-0091
Hung-Chen Chen https://orcid.org/0009-0008-6497-8899
Enoch Yi-No Kang https://orcid.org/0000-0002-4903-942X
Hung-Wei Tsai https://orcid.org/0000-0002-8417-2461
Yi-Chun Chen https://orcid.org/0000-0001-5244-3282
Hao-Yu Chen https://orcid.org/0009-0006-9531-3991
Hui-Wen Chen https://orcid.org/0000-0001-9304-9396
S. Barry Issenberg https://orcid.org/0000-0002-2524-4736

REFERENCES

1. Mohsin SU, Ibrahim Y, Levine D. Teaching medical students to recognise and report errors. *BMJ Open Qual*. 2019;8(2):e000558. doi:10.1136/bmjoq-2018-000558
2. Swinfen D, Labuschagne M, Joubert G. Disclosing medical errors: how do we prepare our students? *BMC Med Educ*. 2023;23(1):191. doi:10.1186/s12909-023-04125-3
3. Kohn, L, Corrigan, J, Donaldson, M. To err is human: building a safer health system, 2000.
4. Slawomirski L, Klazinga N. *The Economics of Patient Safety: From Analysis to Action*. OECD Publishing; 2022.
5. Panagioti M, Khan K, Keers RN, et al. Prevalence, severity, and nature of preventable patient harm across medical care settings: systematic review and meta-analysis. *BMJ*. 2019;366:l4185. doi:10.1136/bmj.l4185
6. Mueller PS. Teaching and assessing professionalism in medical learners and practicing physicians. *Rambam Maimonides Med J*. 2015;6(2):e0011. doi:10.5041/RMMJ.10195
7. Tucker CR, Choby BA, Moore A, et al. Speaking up: using OSTEs to understand how medical students address professionalism lapses. *Med Educ Online*. 2016;21(1):32610. doi:10.3402/meo.v21.32610
8. Broeckelman-Post MA, Norander S, Ball TC, et al. What communication skills do other disciplines value most? A communication across the curriculum needs analysis. *Commun Educ*. 2023;72(1):40-60. doi:10.1080/03634523.2022.2136395
9. Lester H, Tritter JQ. Medical error: a discussion of the medical construction of error and suggestions for reforms of medical education to decrease error. *Med Educ*. 2001;35(9):855-861. doi:10.1046/j.1365-2923.2001.01003.x
10. Wu J, Olagunju AT. Mentorship in medical education: reflections on the importance of both unofficial and official mentorship programs. *BMC Med Educ*. 2024;24(1):1233. doi:10.1186/s12909-024-06248-7
11. Kim S, Appelbaum NP, Baker N, et al. Patient safety over power hierarchy: a scoping review of healthcare professionals' speaking-up skills training. *J Healthc Qual*. 2020;42(5):249-263. doi:10.1097/JHQ.0000000000000257
12. Okuyama A, Wagner C, Bijnen B. Speaking up for patient safety by hospital-based health care professionals: a literature review. *BMC Health Serv Res*. 2014;14:61. doi:10.1186/1472-6963-14-61
13. Chen YC, Issenberg SB, Issenberg Z, Chen HW, Kang YN, Wu JC. Factors associated with medical students speaking-up about medical errors: a cross-sectional study. *Med Teach*. 2022;44(1):38-44. doi:10.1080/0142159X.2021.1959904
14. Peadon R, Hurley J, Hutchinson M. Hierarchy and medical error: speaking up when witnessing an error. *Saf Sci*. 2020;125:104648. doi:10.1016/j.ssci.2020.104648

13652923, 0, Downloaded from https://asmepublications.onlinelibrary.wiley.com/doi/10.1111/medu.70030 by Lendis Wu - Taipei Medical University(Tmu) , Wiley Online Library on [10/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

15. Dybdal Kayser J, Kjær Ersbøll A, Kolbe M, Østergaard D, Dieckmann P. Medical students' speak-up barriers: a randomized controlled trial with written vignettes. *J Patient Saf*. 2024;20(5):323-329. doi:10.1097/PTS.0000000000001227
16. Etchegaray JM, Ottosen MJ, Dancsak T, Thomas EJ. Barriers to speaking up about patient safety concerns. *J Patient Saf*. 2020;16(4):e230-e234. doi:10.1097/PTS.0000000000000334
17. Lee SE, Dahinten VS, Ji H, Kim E, Lee H. Motivators and inhibitors of nurses' speaking up behaviours: a descriptive qualitative study. *J Adv Nurs*. 2022;78(10):3398-3408. doi:10.1111/jan.15343
18. Barzallo Salazar MJ, Minkoff H, Bayya J, et al. Influence of surgeon behavior on trainee willingness to speak up: a randomized controlled trial. *J Am Coll Surg*. 2014;219(5):1001-1007. doi:10.1016/j.jamcollsurg.2014.07.933
19. Omura M, Maguire J, Levett-Jones T, Stone TE. The effectiveness of assertiveness communication training programs for healthcare professionals and students: A systematic review. *Int J Nurs Stud*. 2017;76:120-128. doi:10.1016/j.ijnurstu.2017.09.001
20. Chen YC, Issenberg SB, Chiu YJ, et al. Exploration of students' reaction in medical error events and the impact of personalized training on the speaking-up behavior in medical error events. *Med Teach*. 2023;45(4):368-374.
21. Behmanesh D, Jalilian S, Heydarabadi AB, Ahmadi M, Khajeali N. The impact of hidden curriculum factors on professional adaptability. *BMC Med Educ*. 2025;25(1):186. doi:10.1186/s12909-024-06618-1
22. Brown MEL, Coker O, Heybourne A, Finn GM. Exploring the hidden curriculum's impact on medical students: professionalism, identity formation and the need for transparency. *Med Sci Educ*. 2020;30(3):1107-1121. doi:10.1007/s40670-020-01021-z
23. Wyatt TR, de Oliveira Vidal EI. The social construction of time and its influence on medical education. *Med Educ*. 2025;59(1):97-103. doi:10.1111/medu.15472
24. Hafferty FW. Beyond curriculum reform: confronting medicine's hidden curriculum. *Acad Med*. 1998;73(4):403-407. doi:10.1097/00001888-199804000-00013
25. Hafferty FW, Franks R. The hidden curriculum, ethics teaching, and the structure of medical education. *Acad Med*. 1994;69(11):861-871.
26. Murakami M, Kawabata H, Maezawa M. The perception of the hidden curriculum on medical education: an exploratory study. *Asia Pac Fam Med*. 2009;8(1):9. doi:10.1186/1447-056X-8-9
27. Mahood SC. Medical education: beware the hidden curriculum. *Can Fam Physician*. 2011;57(9):983-985.
28. Torralba KD, Jose D, Byrne J. Psychological safety, the hidden curriculum, and ambiguity in medicine. *Clin Rheumatol*. 2020;39(3):667-671.
29. Farooq M, Rahma AT, Alam Z, et al. Transforming professionalism education in clerkships: a student-driven approach utilizing the hidden curriculum. *Perspect Med Educ*. 2025;14(1):149-161. doi:10.5334/pme.1572
30. Lempp H, Seale C. The hidden curriculum in undergraduate medical education: qualitative study of medical students' perceptions of teaching. *BMJ*. 2004;329(7469):770-773.
31. Noort MC, Reader TW, Gillespie A. Speaking up to prevent harm: a systematic review of the safety voice literature. *Saf Sci*. 2019;117:375-387. doi:10.1016/j.ssci.2019.04.039
32. Giorgi A. The descriptive phenomenological method in psychology. *J Phenomenol Psychol*. 2012;43:3-12.
33. Braun V, Clarke V. Toward good practice in thematic analysis: avoiding common problems and be (com)ing a knowing researcher. *Int J Transgend Health*. 2023;24(1):1-6. doi:10.1080/26895269.2022.2129597
34. Etikan I. Comparison of convenience sampling and purposive sampling. *Am J Theor Appl Stat*. 2016;5(1):1. doi:10.11648/j.ajtas.20160501.11
35. Page RL, Joglar JA, Caldwell MA, et al. 2015 ACC/AHA/HRS guideline for the management of adult patients with supraventricular tachycardia: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines and the Heart Rhythm Society. *Heart Rhythm*. 2016;13(4):e136-e221. doi:10.1016/j.hrthm.2015.09.019
36. Braun V, Clarke V. Using thematic analysis in psychology. *Qual Res Psychol*. 2006;3(2):77-101. doi:10.1191/1478088706qp063oa
37. Glaser BG, Strauss AL. *The Discovery of Grounded Theory: Strategies for Qualitative Research*. Aldine Publishing Company; 1967.
38. Malterud K, Siersma VD, Guassora AD. Sample size in qualitative interview studies: guided by information power. *Qual Health Res*. 2016;26(13):1753-1760. doi:10.1177/1049732315617444
39. Varpio L, Ajjawi R, Monrouxe LV, O'Brien BC, Rees CE. Shedding the cobra effect: problematising thematic emergence, triangulation, saturation and member checking. *Med Educ*. 2017;51(1):40-50. doi:10.1111/medu.13124
40. Lee S. Concept and development of resident training program for general competencies. *Korean Med Educ Rev*. 2017;19(2):63-69. doi:10.17496/kmer.2017.19.2.63
41. Sweigart JR, Tad YD, Kneeland P, Williams MV, Glasheen JJ. Hospital medicine resident training tracks: developing the hospital medicine pipeline. *J Hosp Med*. 2017;12(3):173-176.
42. Schwappach D, Sendlhofer G, Kamolz LP, Köle W, Brunner G. Speaking up culture of medical students within an academic teaching hospital: need of faculty working in patient safety. *PLoS One*. 2019;14(9):e0222461. doi:10.1371/journal.pone.0222461
43. Mira JJ, Matarredona V, Tella S, et al. Unveiling the hidden struggle of healthcare students as second victims through a systematic review. *BMC Med Educ*. 2024;24(1):378. doi:10.1186/s12909-024-05336-y
44. Ozeke O, Ozeke V, Coskun O, Budakoglu II. Second victims in health care: current perspectives. *Adv Med Educ Pract*. 2019;10:593-603. doi:10.2147/AMEP.S185912
45. Jones A, Blake J, Adams M, Kelly D, Mannion R, Maben J. Interventions promoting employee "speaking-up" within healthcare workplaces: a systematic narrative review of the international literature. *Health Policy*. 2021;125(3):375-384. doi:10.1016/j.healthpol.2020.12.016
46. Kossaify A, Hleihel W, Lahoud JC. Team-based efforts to improve quality of care, the fundamental role of ethics, and the responsibility of health managers: monitoring and management strategies to enhance teamwork. *Public Health*. 2017;153:91-98. doi:10.1016/j.puhe.2017.08.007
47. Abbasi M, Shirazi M, Torkmandi H, Homayoon S, Abdi M. Impact of teaching, learning, and assessment of medical law on cognitive, affective and psychomotor skills of medical students: a systematic review. *BMC Med Educ*. 2023;23(1):703. doi:10.1186/s12909-023-04695-2
48. Sullivan E, Thampy H, Gay S. Raising professionalism concerns as a medical student: damned if they do, damned if they don't? *BMC Med Educ*. 2024;24(1):208. doi:10.1186/s12909-024-05144-4
49. Dodsworth A, Munro K, Alberti H, Hirsh DA, Paes P, Illing J. Patient outcomes in a longitudinal integrated clerkship: a systematic literature review. *Med Educ*. 2023;57(9):820-832. doi:10.1111/medu.15013
50. Walters L, Greenhill J, Richards J, et al. Outcomes of longitudinal integrated clinical placements for students, clinicians and society. *Med Educ*. 2012;46(11):1028-1041. doi:10.1111/j.1365-2923.2012.04331.x
51. Teherani A, Irby DM, Loeser H. Outcomes of different clerkship models: longitudinal integrated, hybrid, and block. *Acad Med*. 2013;88(1):35-43. doi:10.1097/ACM.0b013e318276ca9b
52. Orsmond P, McMillan H, Zvauya R. It's how we practice that matters: professional identity formation and legitimate peripheral participation

13652923, 0, Downloaded from https://asmepublications.onlinelibrary.wiley.com/doi/10.1111/medu.70030 by Lendis Wu - Taipei Medical University(Tmu) , Wiley Online Library on [10/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

in medical students: a qualitative study. *BMC Med Educ*. 2022;22(1): 91. doi:10.1186/s12909-022-03107-1

53. Gonzalo JD, Wolpaw D, Graaf D, Thompson BM. Educating patient-centered, systems-aware physicians: a qualitative analysis of medical student perceptions of value-added clinical systems learning roles. *BMC Med Educ*. 2018;18(1):248. doi:10.1186/s12909-018-1345-5

How to cite this article: Wu J-C, Chen H-C, Kang EY-N, et al. Breaking the silence: Revealing drivers and barriers to medical students' speaking up in medical error. *Med Educ*. 2025;1-11. doi:10.1111/medu.70030

13652923, 0, Downloaded from https://asmpublications.onlinelibrary.wiley.com/doi/10.1111/medu.70030 by Lendis Wu - Taipei Medical University(Tmu) , Wiley Online Library on [10/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License